

VisionInspect AI

AI-Powered Manufacturing Defect Detection & Quality Inspection System

System Architecture & Platform Overview Documentation

PatchCore Anomaly Detection · ResNet-18 · FastAPI · React

Prepared for Final Year Project Documentation, Internship Submission,
Project Review & Mentor Presentation

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1. Introduction & Problem Statement

Traditional supervised defect-classification models require large, carefully labeled datasets of both good and defective products — something most manufacturing lines simply do not have, especially for new or low-volume product runs. Collecting and labeling thousands of examples of every possible defect type before a system can even be trained is slow, expensive, and often impractical on a live factory floor.

This document provides a system-level overview of VisionInspect AI: the anomaly-detection methodology it is built on, the technology stack powering the backend and frontend, the platform's core features, and the business case for deploying it on a production line.

2. AI Methodology — Unsupervised Anomaly Detection

To solve the data-scarcity problem, VisionInspect AI abandons traditional classification in favor of Unsupervised Anomaly Detection, using a PatchCore-style approach built on a pre-trained convolutional backbone.

2.1 The Backbone: ResNet-18

The system uses a pre-trained ResNet-18 model as a feature extractor. Because it is pre-trained on a large general-purpose image dataset, it already understands basic shapes, edges, and textures out-of-the-box — no manufacturing-specific training is required before the pipeline can begin learning a product line.

2.2 Training Phase — The Memory Bank

Instead of training on defects, the model is fed only images of perfect, normal products. The ResNet-18 backbone extracts intermediate feature maps (patches) from these good images and stores them in a highly optimized, searchable Memory Bank. No defective samples are ever required to bootstrap a new product line.

2.3 Inference Phase — Defect Detection

When a new product is photographed on the assembly line, the model extracts its own feature patches and uses K-Nearest Neighbors (KNN) search to find the closest matching "normal" patch stored in the Memory Bank.

2.4 Anomaly Scoring

If the distance between a new patch and its closest normal match exceeds a defined threshold, that patch is flagged as an anomaly. The aggregated patch-level distances produce the image-level confidence and severity scores (Low, Medium, High, Critical) surfaced throughout the platform.

2.5 Visual Explainability — Heatmaps

Because the model tracks exactly which pixel patches deviated from the norm, it generates a high-resolution Heatmap Mask. This mask highlights the precise location and shape of the scratch, dent, contamination, or discoloration, and is rendered side-by-side with the original image in the UI so operators can visually verify every AI decision rather than trusting a black-box label.

3. System Architecture & Tech Stack

The architecture is decoupled into a modern frontend and a scalable microservice-style backend, connected over a JSON REST API.

3.1 Backend Infrastructure (FastAPI & Python)

- **FastAPI Engine:** Chosen for its asynchronous capabilities, allowing the server to handle heavy image processing and AI inference without blocking other requests.
- **PostgreSQL & SQLAlchemy:** A relational database schema tracks Users, Roles, Inspection Logs, and Image Metadata — including file paths for both the original image and the generated heatmap.
- **Security (JWT & RBAC):** Stateless JSON Web Tokens handle authentication. Role-Based Access Control ensures that only admin and quality_engineer roles can trigger heavy AI inference, while factory_supervisor roles are restricted to viewing analytics.

3.2 Frontend Architecture (React & Tailwind)

- **Vite + React:** Ensures lightning-fast Hot Module Replacement during development and highly optimized production builds.
- **State Management & Routing:** Handled via React Router with protected route wrappers (<ProtectedRoute>, <RoleGuard>) that instantly redirect unauthorized users.
- **UI/UX — Neon Cyber Aesthetic:** The interface is designed to feel like next-generation industrial software. It uses a #0a0a0f base background with glowing magenta-to-cyan gradients, glassmorphism panels (backdrop-blur), and custom micro-animations (Framer Motion) to reduce cognitive load on operators.
- **PDF Export Module:** Integrated html2canvas and jsPDF to let operators capture the DOM elements of an inspection result and compile them into a downloadable, shareable QA report without requiring a backend PDF-generation service.

3.3 Technology Stack Summary

Layer	Technology	Purpose
AI / Modeling	ResNet-18 + PatchCore (KNN)	Unsupervised feature extraction & anomaly scoring
Backend API	FastAPI (Python, async)	Auth, image intake, inference orchestration
Database	PostgreSQL + SQLAlchemy ORM	Users, roles, inspection logs, image metadata
Auth & Access	JWT + Role-Based Access Control	Stateless auth; admin / quality_engineer / factory_supervisor
Frontend	React (Vite) + Tailwind CSS	SPA UI, protected routing, dashboards
Motion / UX	Framer Motion	Micro-animations for the neon cyber

Layer	Technology	Purpose
		interface
Reporting	html2canvas + jsPDF	Client-side, backend-free QA report export
History Grid	AG Grid	Searchable, sortable inspection audit trail

4. Key Platform Features

- **Role-Based Dashboards:** A dynamic UI that adapts to the user. Supervisors see real-time metrics (Yield Rate, Average Confidence, Inspection Volume Area Charts), while Engineers see actionable upload zones.
- **Drag-and-Drop Inference:** A stylized upload zone that supports immediate image submission to the FastAPI backend, automatically transitioning the user to the analysis view upon completion.
- **Side-by-Side Analysis View:** The /results page renders the original high-resolution image adjacent to the AI Heatmap, complete with Defect Severity Scoring (Low, Medium, High, Critical) based on the mathematical deviation of the anomaly.
- **Historical Audit Trail:** An AG Grid-powered history table logs every inspection, displaying thumbnail previews of heatmaps so engineers can rapidly audit past production runs.

5. Platform Walkthrough (Screens)

The following screens illustrate the platform in operation, from an operator's first login through defect analysis, historical audit, analytics, and administration.

5.1 Factory Overview Dashboard

On login, operators land on the Factory Overview dashboard, showing headline metrics — Total Inspections, Healthy Products, Defective Products, and Yield Target — alongside an Inspection Volume Trend chart and live System Health status for the API Gateway, AI Model Nodes, and Database Sync.

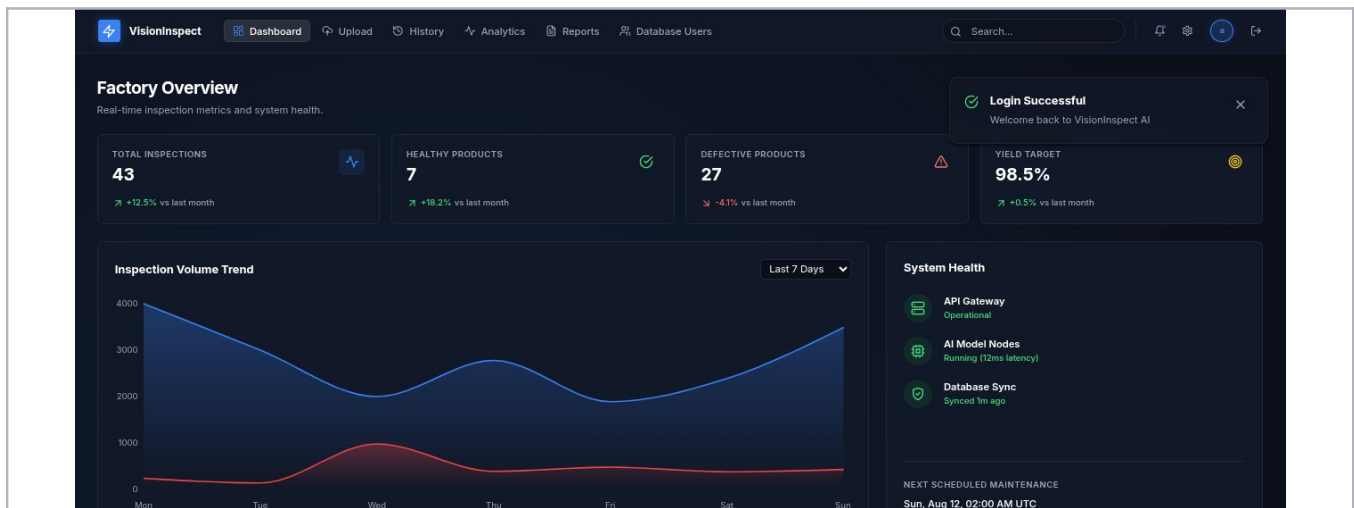


Figure 1: Factory Overview dashboard with real-time inspection metrics and system health.

5.2 New Inspection — Drag-and-Drop Upload

The Upload screen lets an operator select a Product Category (Bottle, Cable, Capsule, Carpet, Grid, Hazelnut, Leather, Metal Nut, Pill, Screw, Tile, Toothbrush, Transistor, and more) and drag-and-drop or browse for an image before running AI Inspection.

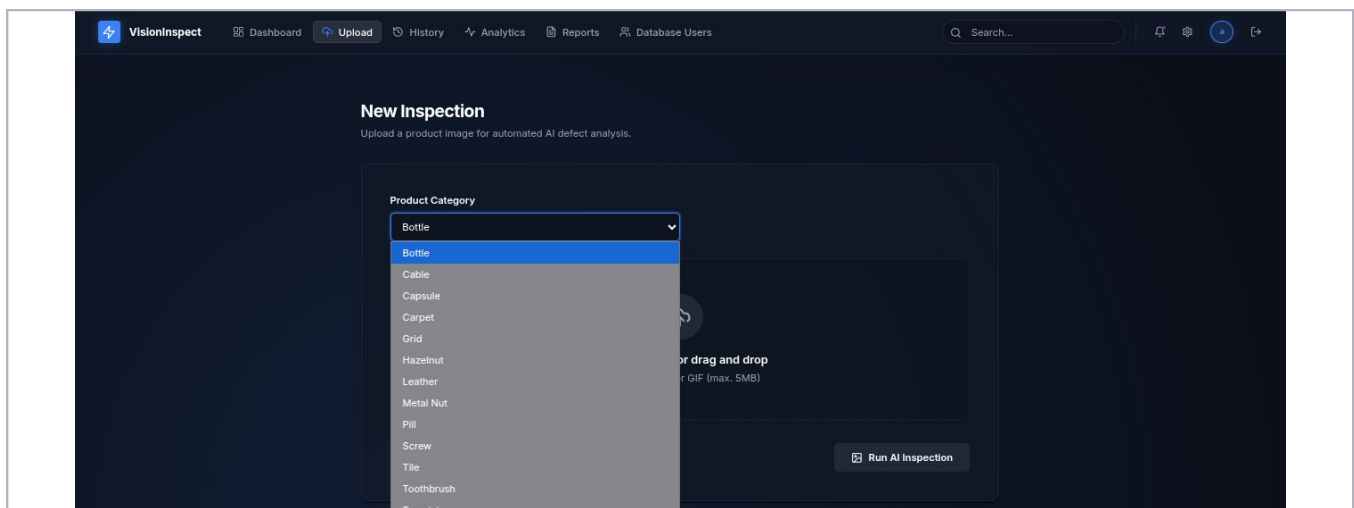


Figure 2: New Inspection screen with product-category selector and drag-and-drop upload zone.

5.3 Inspection Report — Side-by-Side Heatmap Analysis

Once inference completes, the Inspection Report renders the Original Image beside the AI Heatmap Analysis, along with Confidence, Severity, Defect Type, Model Used, Affected Area, and Location Score — giving the operator a fully explainable result rather than a bare label.

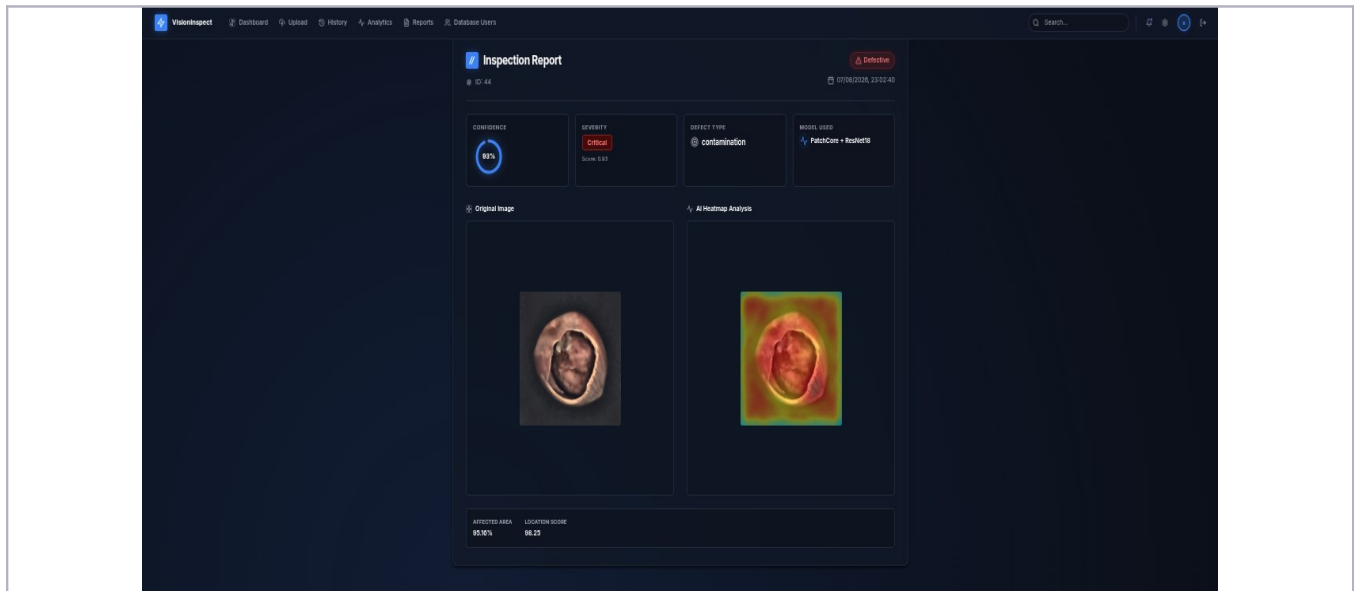
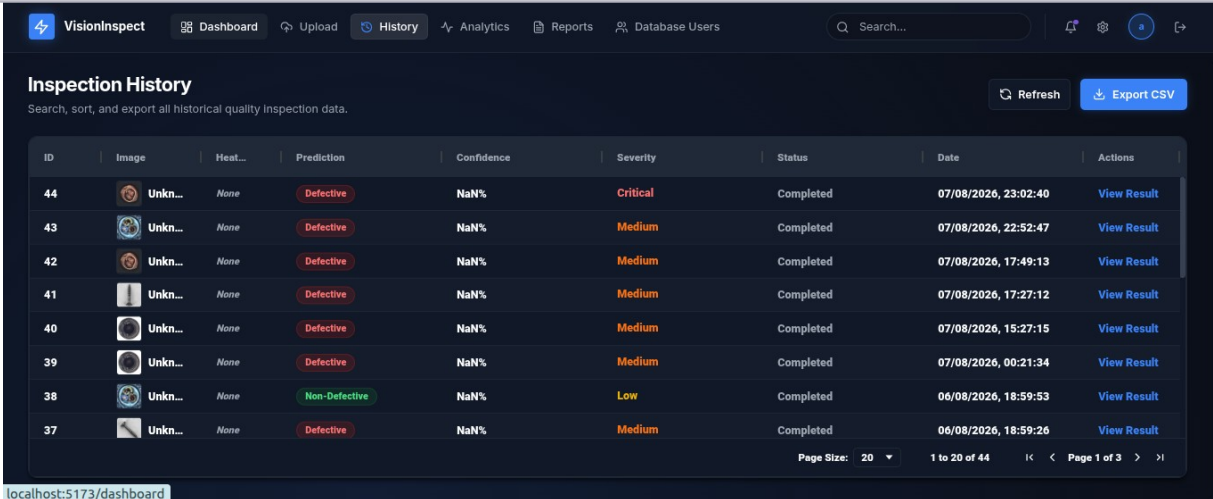


Figure 3: Inspection Report showing a Critical-severity contamination defect with confidence and heatmap overlay.

5.4 Inspection History & Audit Trail

The History screen presents a searchable, sortable, CSV-exportable table of every inspection ever run, including prediction, confidence, severity, status, and a direct link to view each full result — supporting traceability and compliance audits.



The screenshot shows the 'Inspection History' interface. At the top, there's a navigation bar with 'VisionInspect', 'Dashboard', 'Upload', 'History' (active), 'Analytics', 'Reports', and 'Database Users'. Below the navigation bar, the 'Inspection History' title is followed by a subtitle 'Search, sort, and export all historical quality inspection data.' and buttons for 'Refresh' and 'Export CSV'. The main content is a table with the following columns: ID, Image, Heat..., Prediction, Confidence, Severity, Status, Date, and Actions. The table contains 8 rows of data, each with a 'View Result' link in the Actions column. The bottom of the table shows 'Page Size: 20' and '1 to 20 of 44'.

ID	Image	Heat...	Prediction	Confidence	Severity	Status	Date	Actions
44	Unkn...	None	Defective	NaN%	Critical	Completed	07/08/2026, 23:02:40	View Result
43	Unkn...	None	Defective	NaN%	Medium	Completed	07/08/2026, 22:52:47	View Result
42	Unkn...	None	Defective	NaN%	Medium	Completed	07/08/2026, 17:49:13	View Result
41	Unkn...	None	Defective	NaN%	Medium	Completed	07/08/2026, 17:27:12	View Result
40	Unkn...	None	Defective	NaN%	Medium	Completed	07/08/2026, 15:27:15	View Result
39	Unkn...	None	Defective	NaN%	Medium	Completed	07/08/2026, 00:21:34	View Result
38	Unkn...	None	Non-Defective	NaN%	Low	Completed	06/08/2026, 18:59:53	View Result
37	Unkn...	None	Defective	NaN%	Medium	Completed	06/08/2026, 18:59:26	View Result

Page Size: 20 1 to 20 of 44 Page 1 of 3

Figure 4: Inspection History table with per-record severity, status, and result links.

5.5 Quality Analytics

The Analytics screen breaks down defect distribution by type (Broken Large, Broken Small, Contamination, Scratch) and category performance (Passed vs. Failed) across product categories such as Bottle, Cable, Capsule, Metal Nut, Pill, and Transistor, with a one-click Export Report.



Figure 5: Quality Analytics — defect distribution and category performance.

5.6 System Admin Console

Administrators manage operator accounts and roles (Factory Supervisor, Admin, Quality Engineer), active defect categories, and severity-threshold sliders from a dedicated System Admin Console — keeping platform-wide configuration separate from day-to-day inspection work.

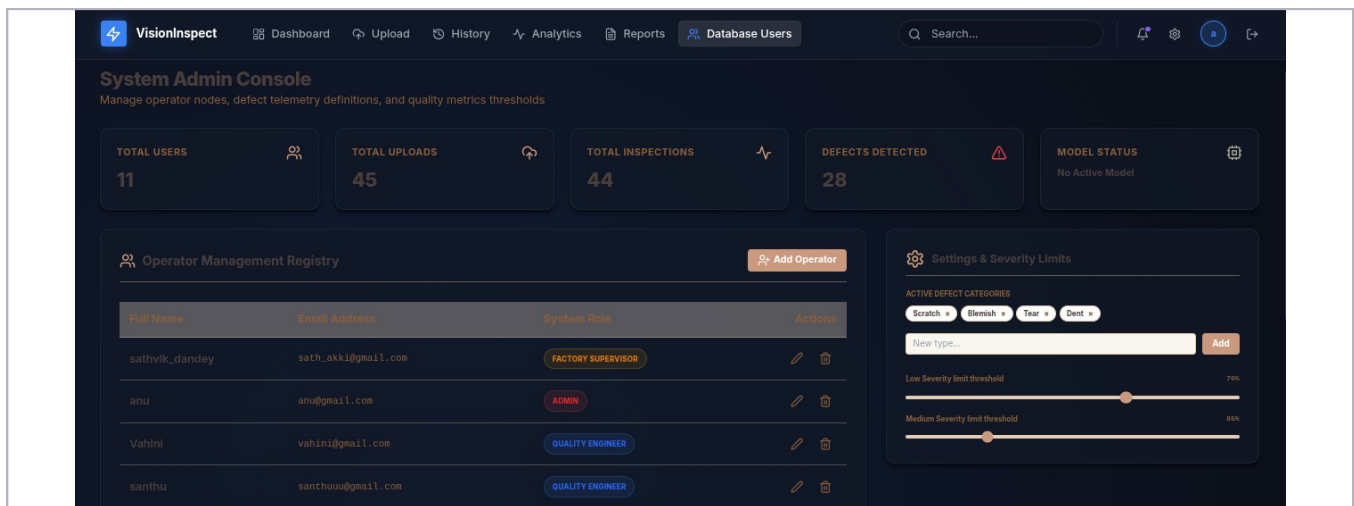


Figure 6: System Admin Console — operator registry and severity-threshold settings.

6. Business Impact & ROI (Return on Investment)

- **Increased Throughput:** Reduces inspection time per unit from seconds/minutes (human) to milliseconds (AI).
- **Zero-Defect Manufacturing:** Prevents defective units from reaching consumers, drastically reducing warranty claims and product recalls.
- **Rapid Deployment:** Because PatchCore only requires "good" images to train, the system can be deployed on a brand-new product line in hours rather than months.

7. Future Scope & Scalability

- **Edge Deployment:** Future iterations could containerize the FastAPI/PatchCore backend using Docker and deploy it directly onto edge devices (e.g. NVIDIA Jetson) on the factory floor for ultra-low-latency, offline inference.
- **Multi-Camera Support:** Expanding the UI to accept and stitch together images from multiple camera angles for 360-degree product inspection.
- **Continuous Learning:** Implementing a feedback loop in the UI where Quality Engineers can mark "False Positives," automatically updating the model's Memory Bank to make the AI smarter over time.